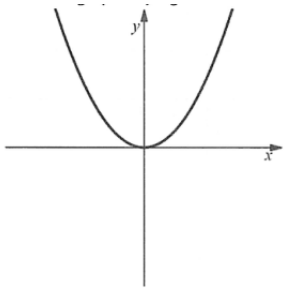
# Proportion

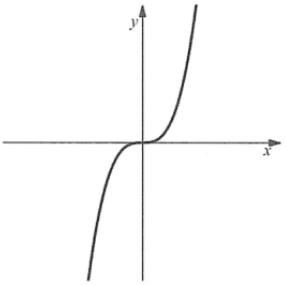
Questions 1–9

**Question 1.** It is given that $y$ is inversely proportional to $x^3$. State which of the following diagrams could represent the graph of $y$ against $x$.

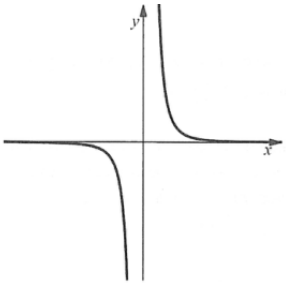
**A**



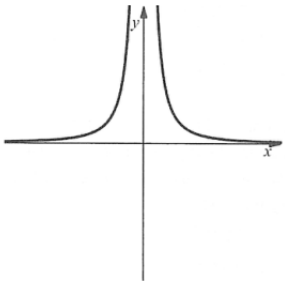

**B**




**C**




**D**




**Question 2.** In each of these cases, write down an equation for $y$ in terms of $x$ and $k$, where $k$ is a constant of proportionality:

a) $y$ is directly proportional to the square of $x$
b) $y$ is inversely proportional to $x$
c) $y$ is directly proportional to the cube root of $x$
d) $y$ is inversely proportional to the cube of $x + 1$

**Question 3.** Given that $y$ is directly proportional to $x^2$, and when $x = 5$, $y = 100$, find $y$ when $x = 10$.

**Question 4.** Given that $y$ is inversely proportional to $\sqrt[4]{x}$, and when $x = 81$, $y = 18$, find $x$ when $y = 180$.

**Question 5.** The volume of gas, $V$ litres, that a gas canister can hold is directly proportional to the temperature, $T$ Kelvin. It is given that when $T = 300$, $V = 24$. Determine the volume of gas when the temperature is 400 Kelvin.

**Question 6.** The time taken, $t$ hours, to complete a job is inversely proportional to the number of workers, $n$, employed. It is given that when $n = 5$, $t = 8$. Determine the number of workers required to complete the job in 4 hours.

**Question 7.** The rate of reaction, $R$ mol/s, of a certain chemical reaction is directly proportional to the concentration $C$ mol/L of a reactant. It is given that when $C = 0.5$, $R = 2$. Determine the concentration that will give a reaction rate of 6 mol/s.

**Question 8.** The intensity of light, $I$ lumens, from a light source is inversely proportional to the square of the distance $d$ metres from the source. It is given that when $d = 2$, $I = 400$. Determine the distance that will give a light intensity of 100 lumens.

**Question 9.** It is given that $y$ is directly proportional to $\sqrt{x}$, and that $z$ is inversely proportional to $y^{10}$. Given that $z = 2$ when $x = 4$, find $x$ when $z = \frac{1}{16}$.